



MACH

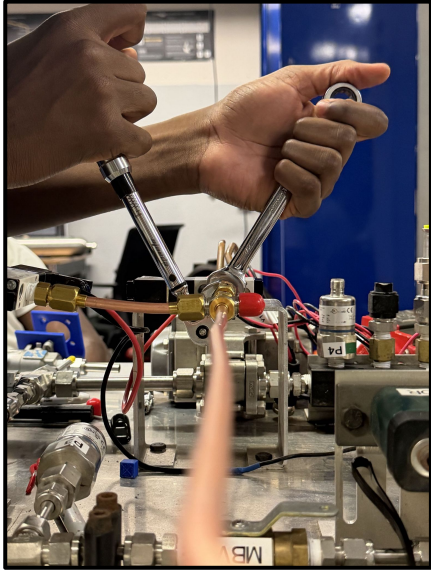
2026 KICKOFF

What We Do

- Design liquid rocket engines and ground-support systems, then put them through real test campaigns.
 - Liquid bi-propellant rocketry
 - Fuel: Ethanol
 - Oxidizer: Nitrous oxide
 - Thrust: 1.3 kN
- Launch Canada Challenge
 - Technology Development Category



Our System



- Development of technology about liquid rocket systems
 - Automated and highly modular system
 - Systems integration
 - New fun projects (torch igniter, TVC, regenerative cooling engine)
 - Pushing boundaries e.g. first student team to demonstrate an engine relight

SERAPHINA

HOTFIRE · AUG 06 2026 · 19:50:48.033 EDT

CAM 03



T-03.03s

SEQUENCE STATE

COUNTDOWN

SUPPLY PRESSURES

N₂
1679 PSI

PNEUMATIC
109 PSI

PURGE
547 PSI

OX FILL
768 PSI

MAIN VALVES

● S1 MAIN OX

CLOSED ● S3 MAIN FUEL

CLOSED

OXIDIZER TANK
0-700 PSI

655 PSI

FUEL TANK
0-700 PSI

645 PSI

CHAMBER
0-280 PSI

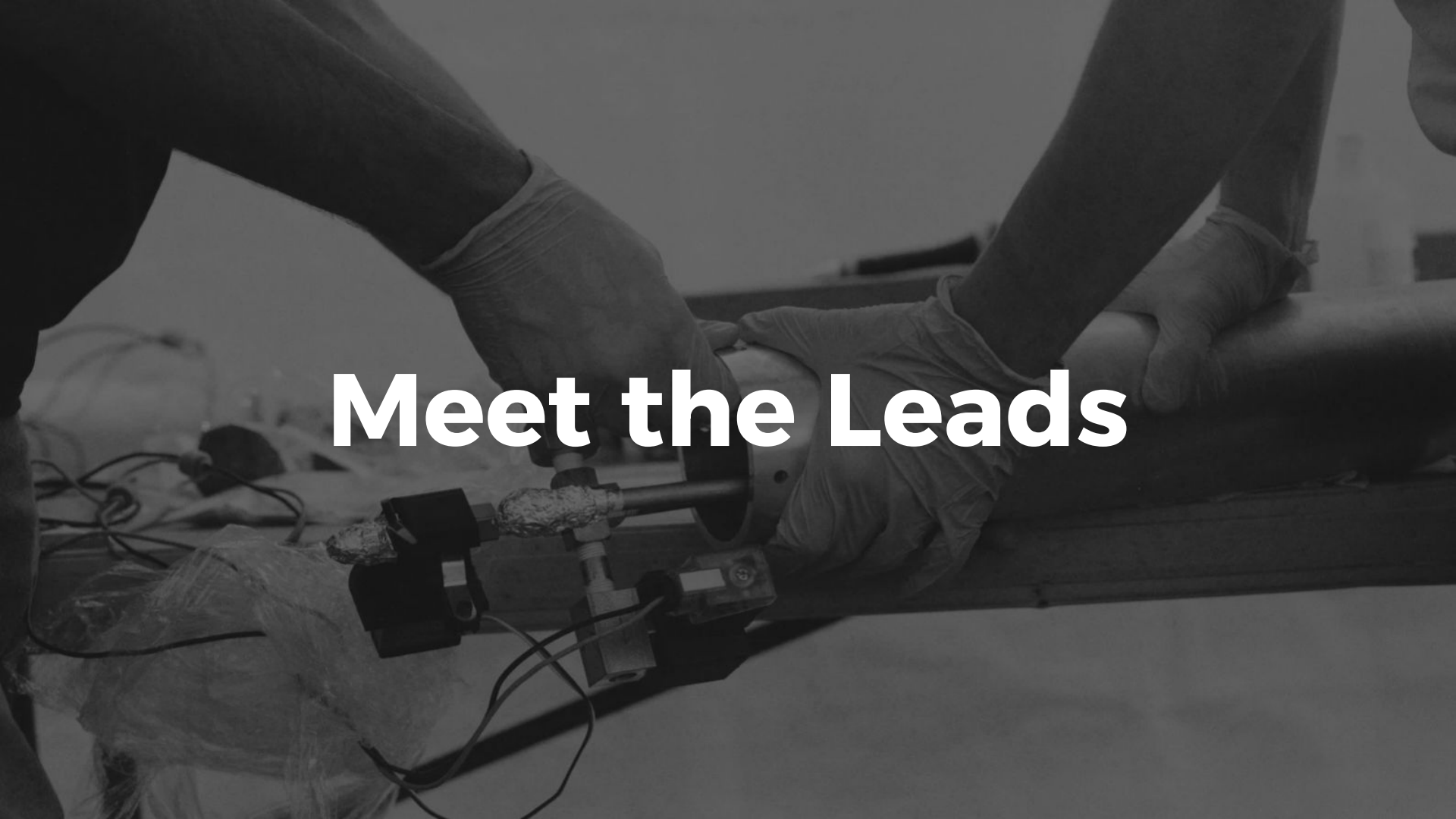
12.3 PSI

THRUST
0-1400 N

1N

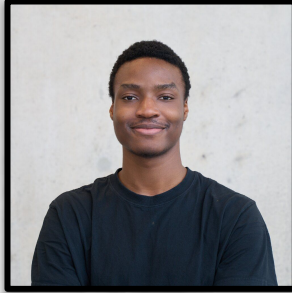
PROPELLANT WEIGHT
0-8 KG

7.1 KG

A black and white photograph showing a person wearing a lab coat and gloves, working on a piece of equipment. The person's hands are visible, one holding a cylindrical component and the other adjusting a knob or switch. The equipment has various wires and components attached to it. The background is slightly blurred, showing what appears to be a laboratory or workshop setting.

Meet the Leads

Propulsion Lead



Tobechukwu Okoh

Team Captain



Julia Puszynska

Operations Director



Samuel Li

Safety Officer



Jonathan Al-hinn

Electrical Lead

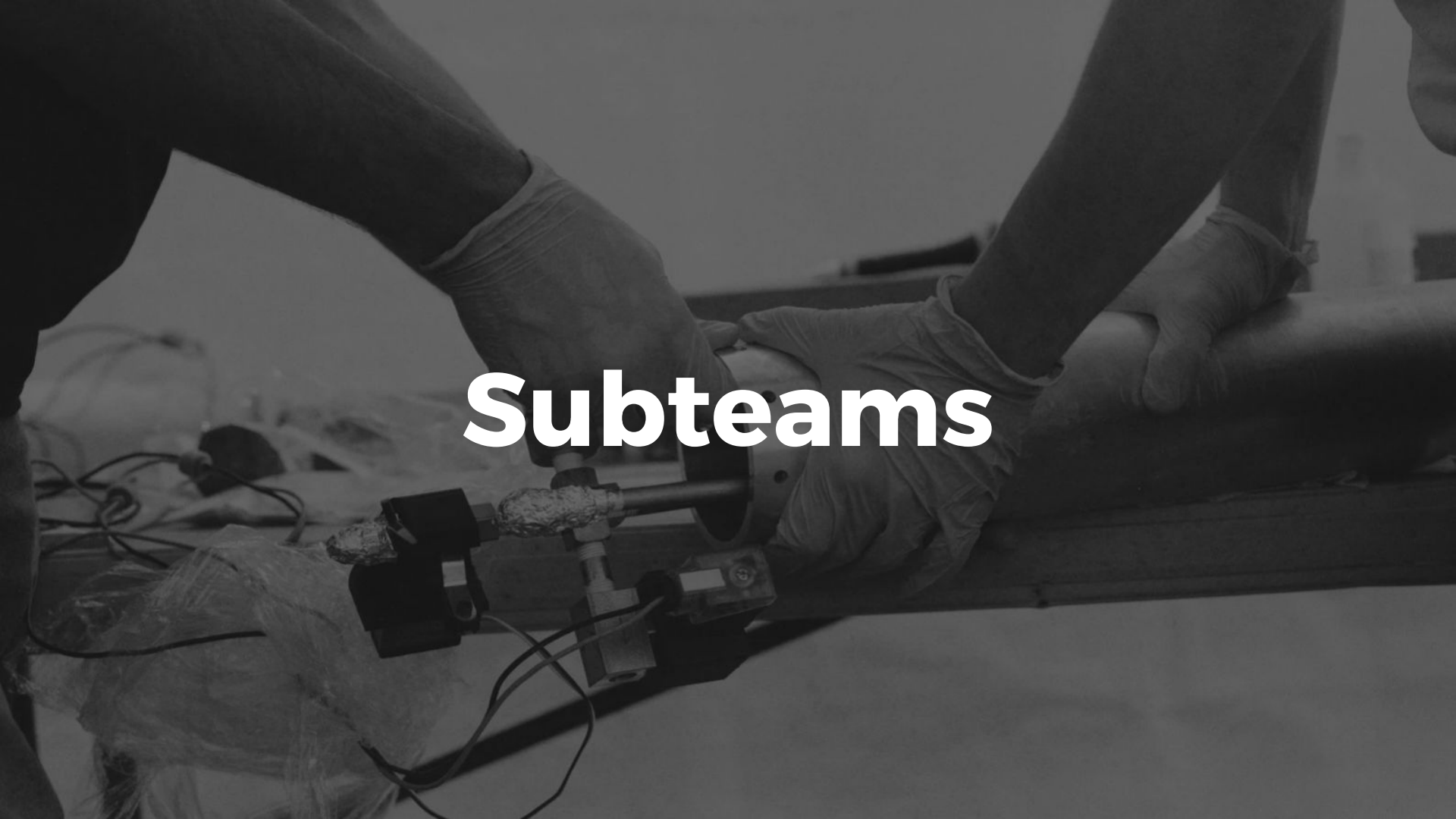


Kasper Pajak

Media & Logistics Lead



Madison Warren

A grayscale photograph showing a person's hands, wearing white gloves, working on a mechanical device. The device appears to be a small, cylindrical component mounted on a larger structure. The person is using a tool, possibly a screwdriver, to adjust or work on the device. The background is slightly blurred, showing some other equipment and a workbench. The word "Subteams" is overlaid in large, white, sans-serif font across the center of the image.

Subteams

Propulsion

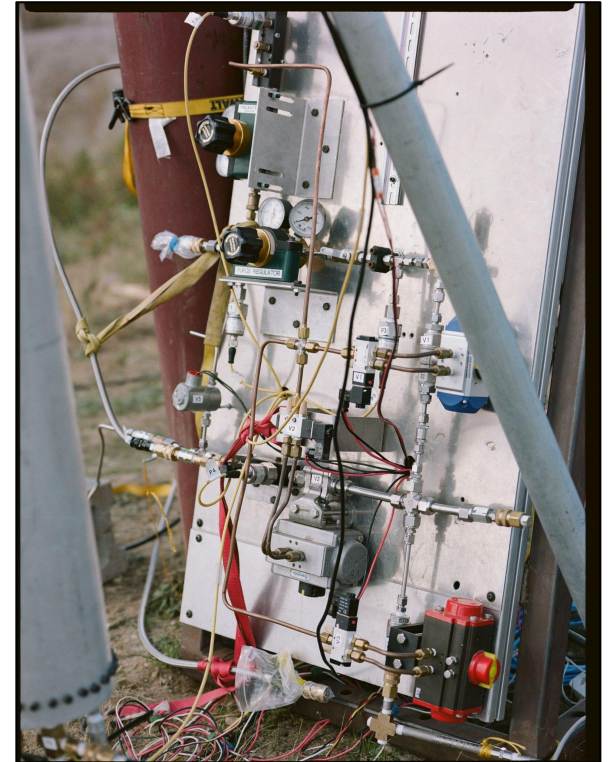
Presented by Tobe

What does propulsion do?

- Design, build, test, and maintain propellant management systems, equipment, and rocket engines.
- Use basic engineering concepts like fluids, pressure, heat, and forces to make design decisions.
- Work with electrical and structures to ensure mechanical, electrical, and fluid interfaces work together.

Minimum Requirements

- Basic engineering knowledge of concepts in fluids, pressure, heat, and forces to make design decisions.
- Willingness to help out with documentation, assembly and general preparations



Electrical & Integration - GSEE

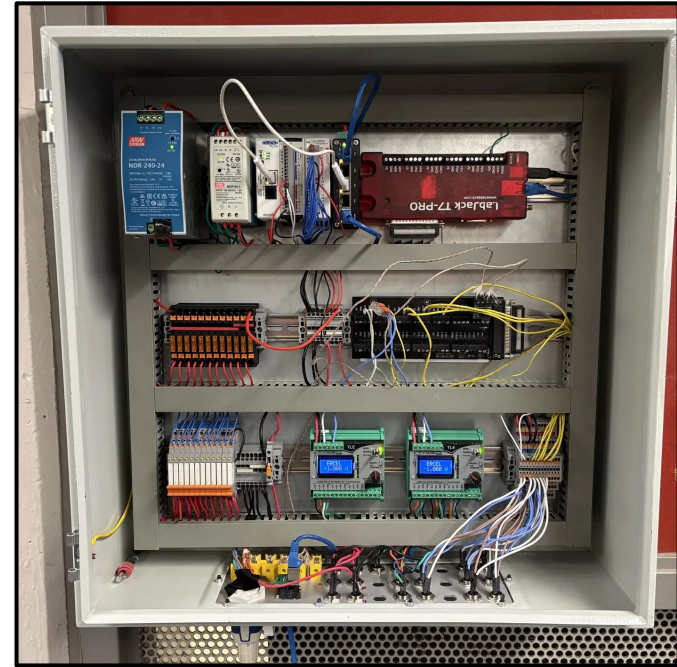
Presented by Kasper

GSEE

- Responsible for developing, maintaining, and operating ground support equipment electronics.
- Integrates various sensors with output devices (valves)
- Basic PLC programming.
- **Focuses on Industrial Automation**

Minimum requirements:

- Knowledge of basic electrical theory (Arduino-based circuits are acceptable)
- Intermediate programming background in any language.



MACH GSEE, SPRINT-era (outdated photo)

Electrical & Integration - Mission Control

Presented by Kasper

Mission Control

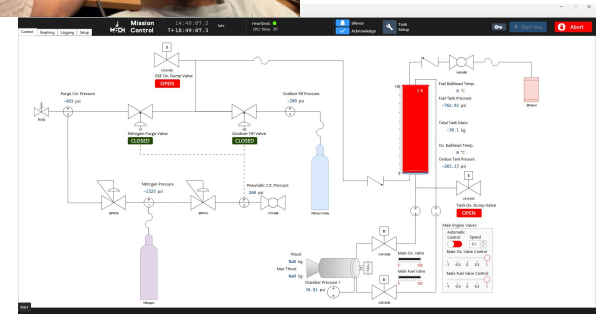
- Oversees all test and flight operations.
- Responsible for ensuring systems operate safely and within specifications.
- Occasionally develops mission control software using LabVIEW

Minimum requirements:

- Must be able to work and remain focused and professional under pressure.
- Must be able to speak clearly.
- Proficient in basic data management.

Useful but not required:

- Prior experience working with LabVIEW.
- Prior experience with radio communications (ROC-A/M/MC, HAM license).



SPRINT Test Control, SOP configuration

Structures

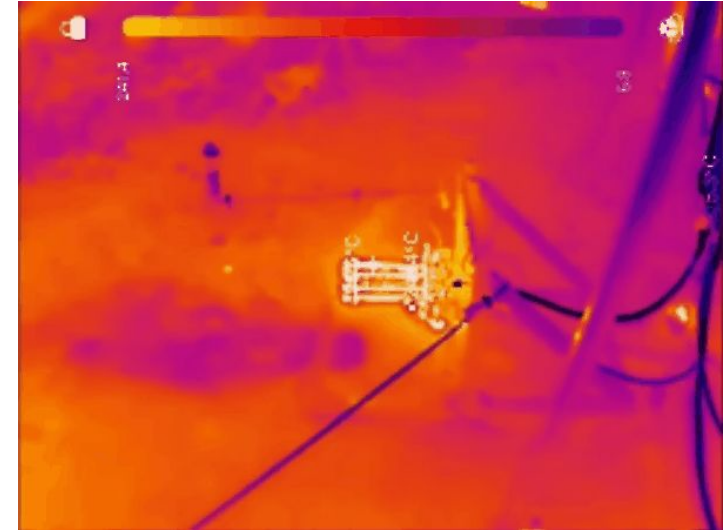
Presented by Tobe

What does structures do?

- Design the parts that support and hold together the rocket and test-system hardware.
- Make sure components can safely handle forces, pressure, vibration, and weight.
- Create models and drawings for structural parts and assemblies.

Minimum Requirements

- Basic knowledge in CAD (Computer-Aided Design) or FEA (Finite Element Analysis)
- Basic understanding of structural concepts (i.e stress)
- Willingness to help out with documentation, assembly and general preparations



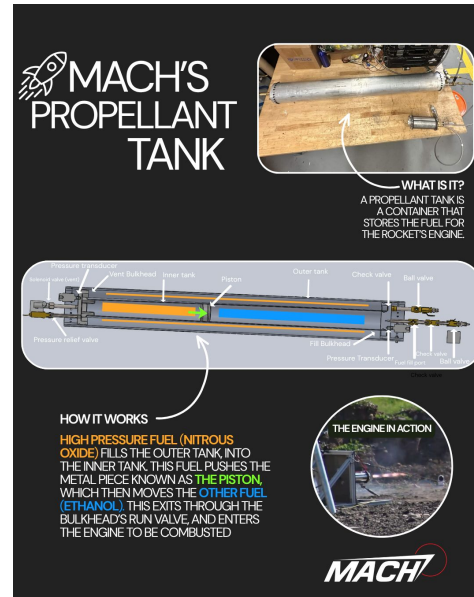
Media

Presented by Madison

What does media do?

Social Media + Content Creation

- Communications with other rocketry teams (collaboration, interactions, reposts)
- Progress updates, events, general information posted to socials
- Merch creation (hoodies, flight tags, etc)



Logistics/Administration

Presented by Sam

What does logistics/administration do?

- Managing all team assets
- Representation and outreach assistance
- Sponsorships

What you learn

- Project Management
- Valuable team soft skills



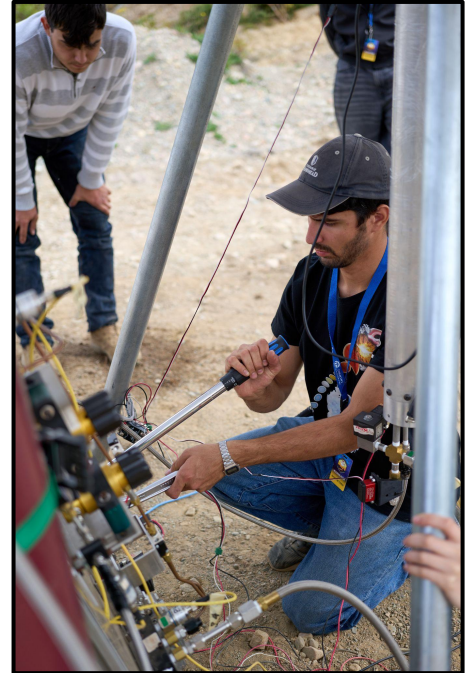
A grayscale photograph showing a person's hands, wearing white gloves, working on a mechanical device. The device appears to be a prosthetic arm or a similar mechanical assembly, with various components like a motor, gears, and wires visible. The person is holding the device, and their hands are positioned to adjust or assemble it. The background is slightly blurred, showing some other equipment or a workshop setting. The text "Learning Opportunities" is overlaid in a large, white, sans-serif font across the center of the image.

Learning Opportunities

Learning Opportunities

Short Term - *these are limited*

- October Hotfire
 - Show up at the general meetings for more info
- Tank Manufacturing
 - Keep an eye out in the discord!
- Engine + Tank Assembly
 - Keep an eye out in the discord!



Learning Opportunities

Intermediate Term

- Gas Cylinder Training
 - Late October
- Engine Design
 - Limited places
- General Subteam Tutorials
 - Propulsion Tutorial
 - Eleclectrical tut.
 - Radio communication tut.
 - More!
 - Keep an eye out in the discord!



Hands-on cylinder training

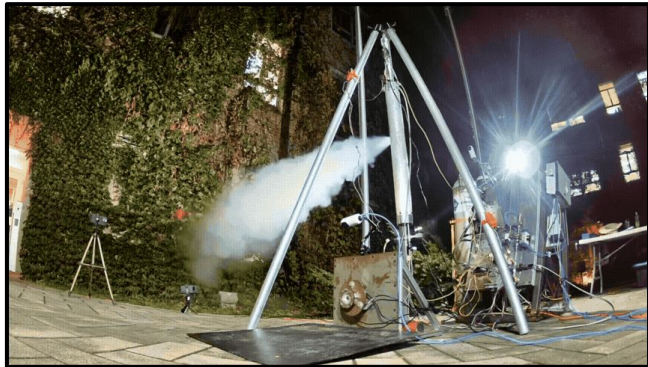
Pressure Tank Manufacturing



Learning Opportunities

Long Term

- Launch Canada
- Coldflow and Hotfire Tests
- Manufacturing



*Coldflow test in
the Quad*



*Launch Canada
Ready!*

Learning Opportunities

- **YOU CHOOSE YOU!**
 - Ex: if you are in electrical but propulsion is doing a really cool project you can be apart of it
 - Mold yourself into what you want to be
 - Learn and grow in your field of interest
 - Research topics that interest you
- **Project Pitching!**
- **Represent MACH at over a dozen different events**
 - Internal and external





Closing Remarks



Who we look for:

- Passionate students that are curious and willing to learn new things
- People who are **team players**
- **Students who aren't afraid to ask questions and admit mistakes**
- Dedicated students willing to consistently make the effort to show up
- We expect you to consider this team like an extra course
 - If you are looking to be an active member

Safety Warnings

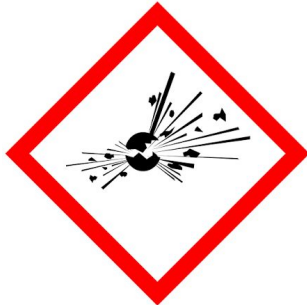
SAFETY IS TOP PRIORITY

- What you need to know
- How we enforce safety
- Do not be afraid
- Your responsibility and expectation



SOME DANGERS INCLUDE BUT ARE NOT LIMITED TO:

- High pressure gasses
- Extreme temperatures
- Explosives
- Shrapnel
- Loud Noises



Thanks for Attending!

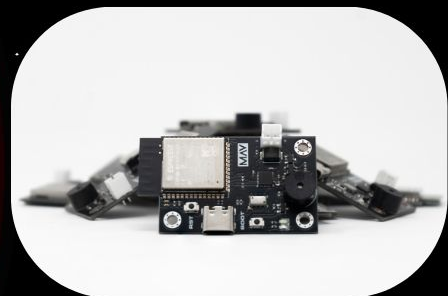


Check out our Linktree:





MARS



INSTAGRAM



Learn how to design and manufacture your own
model rocket with your friends!

F2026 Kick-Off Meeting NEXT WEEK!!

DISCORD





Join our

Discord

